

Experiment No.	3
Experiment Title	Regression Analysis using Linear and Regularized Models ¹
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1 Objective

Implement Linear, Ridge, Lasso and Elastic Net regression, compare them on multiple error metrics, and work through what overfitting, underfitting and the bias-variance tradeoff look like on a real dataset.

2 Problem Statement

Given loan application records with numeric and categorical fields, predict the loan sanction amount in USD, and see how much regularization actually helps once a baseline linear model is already fit.

3 Dataset Description

Dataset Name	Predict Loan Amount Data
Dataset Source	Kaggle, phileinsophos/predict-loan-amount-data
Number of Samples	29322 (after removing rows with a missing or sentinel target)
Number of Features	45 after one-hot encoding 9 categorical columns
Number of Classes	not applicable, regression target
Missing Values	worst columns were Type of Employment (about 24 percent), Property Age (about 16 percent), Income (about 15 percent)
Train-Test Split	80/20, plus 5 fold cross validation for tuning

338 rows had the target recorded as -999 , a placeholder, and 340 rows had no target at all. Both groups were dropped rather than imputed.

4 Exploratory Data Analysis

- Target distribution: heavily right skewed, a large spike near zero.
- Feature vs target scatter for the four numeric fields expected to matter most.
- Correlation and missing value heatmaps across the raw numeric columns.

¹GitHub Repository: <https://github.com/sanjay-6727/Introduction-to-Machine-Learning->

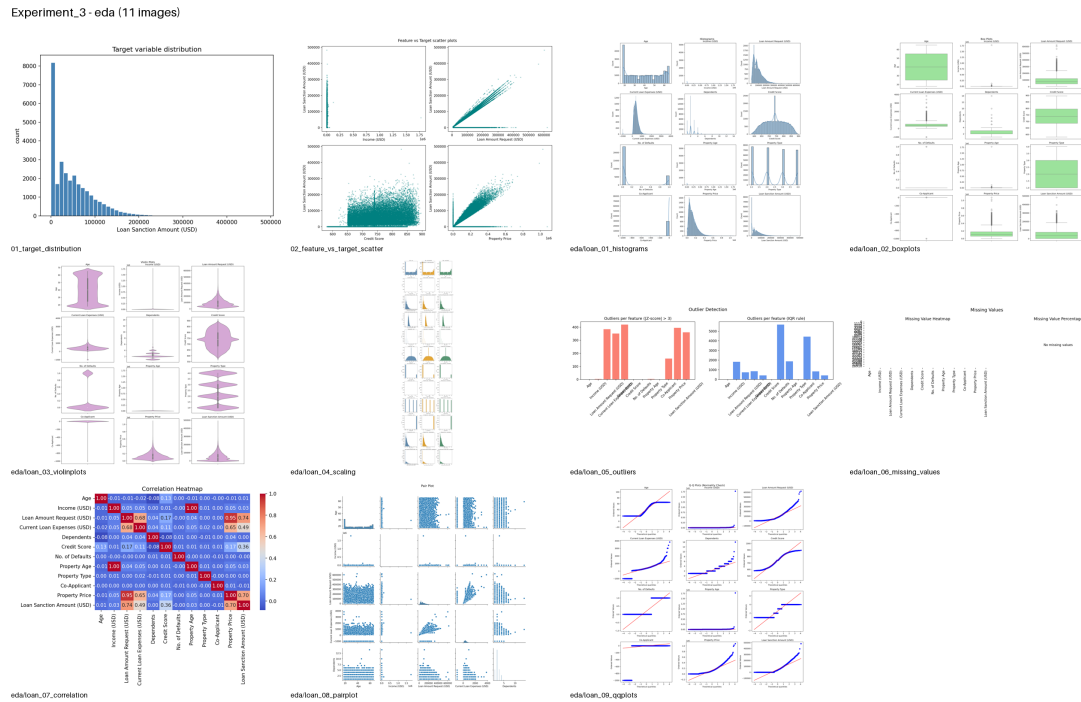


Figure 1: Target distribution, feature vs target scatter plots, and the histogram, box plot, violin, scaling, outlier, missing value, correlation, pair plot and Q-Q panels for the raw numeric features.

Inference:

Loan Amount Request has an almost linear relationship with the sanctioned amount (correlation around 0.74), easily the strongest single predictor. Income and Property Price are far noisier. The missing value heatmap flags Type of Employment, Property Age and Income as the real gaps, and box plots show a handful of extreme outliers in Property Price and Current Loan Expenses.

5 Data Preprocessing

- Numeric columns filled with the median, categorical with the mode. Invalid target rows dropped.
- 9 categorical columns one-hot encoded (first category dropped to avoid the dummy trap).
- All 45 features standardized, important since Ridge/Lasso/Elastic Net penalize coefficient size.
- Identifier columns (Customer ID, Name, Property ID) dropped before modeling.

Listing 1: Grid search over Ridge, Lasso and Elastic Net, 5 fold CV

```

1 param_grids = {
2     "Ridge": (Ridge(), {"alpha": [0.01, 0.1, 1, 10, 100]}),
3     "Lasso": (Lasso(max_iter=50000), {"alpha": [0.001, 0.01, 0.1,
4         1, 10]}),

```

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4     "Elastic Net": (ElasticNet(max_iter=50000),
5         {"alpha": [0.01, 0.1, 1, 10], "l1_ratio": [0.2, 0.5,
6             0.8]}),
7     }
8 for name, (model, grid) in param_grids.items():
9     search = GridSearchCV(model, grid, cv=5, scoring="r2")
10    search.fit(X_train_scaled, y_train)

```

6 Machine Learning Algorithm

Linear Regression fits $y = w^T x + b$ by minimizing squared error, no penalty on the weights.

Ridge Regression adds an L2 penalty, $\|y - Xw\|^2 + \alpha\|w\|_2^2$, shrinking coefficients toward zero without ever hitting exactly zero.

Lasso Regression uses an L1 penalty, $\|y - Xw\|^2 + \alpha\|w\|_1$, which can push some coefficients all the way to zero.

Elastic Net mixes both penalties, aiming for a middle ground between Ridge's stability and Lasso's sparsity, and tends to handle correlated features more gracefully than pure Lasso.

7 Experimental Setup

Python Version	3.11.9
Scikit-Learn Version	1.7.2
IDE	Anaconda Navigator / VS Code
Operating System	Windows 11
Hardware	

8 Hyperparameter Selection

Parameter	Value
Ridge alpha grid	0.01, 0.1, 1, 10, 100
Lasso alpha grid	0.001, 0.01, 0.1, 1, 10
Elastic Net grid	alpha in {0.01,0.1,1,10}, l1_ratio in {0.2,0.5,0.8}

Model	Search Method	Best Parameters	Best CV R2
Ridge	Grid Search	alpha = 100	0.5812
Lasso	Grid Search	alpha = 10	0.5775
Elastic Net	Grid Search	alpha = 0.1, l1_ratio = 0.5	0.5904

9 Results

Cross Validation Performance (K=5)

Model	MAE	MSE	RMSE	R2
Linear Regression	21122.68	9.69e8	31130.38	0.5857
Ridge	21140.84	9.62e8	31015.07	0.5888
Lasso	21113.68	9.65e8	31060.66	0.5876
Elastic Net	21325.89	9.57e8	30930.64	0.5911

Test Set Performance

Model	MAE	RMSE	R2	Search Time (s)
Linear Regression	20760.58	29915.93	0.6021	0.05
Ridge	20782.62	29930.74	0.6017	10.8
Lasso	20751.62	29915.28	0.6021	692.4
Elastic Net	21035.64	30148.24	0.5959	22.9

Search time is the full GridSearchCV run per model, not a single fit, some low alpha Lasso candidates converged slowly.

Coefficient Comparison (three largest in the plain Linear model)

Feature	Linear	Ridge	Lasso	Elastic Net
Property Age	-46130.01	-929.64	-10.53	-88.91
Income (USD)	46116.08	912.03	0.00	71.18
Loan Amount Request	34705.31	33215.74	34503.20	24888.48

10 Result Visualization

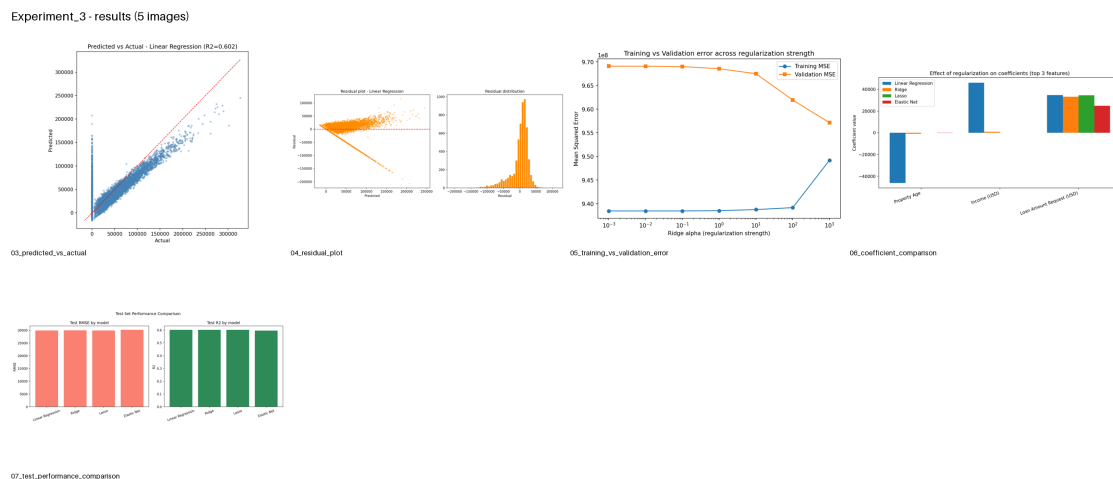


Figure 2: Predicted vs actual scatter, residual plot, training vs validation error across Ridge alpha, coefficient comparison, and test set RMSE/R2 comparison across all four models.

Inference:

Predictions sit close to the diagonal for small to mid loan amounts but the model under-predicts the largest ones, there is simply less data up there. The residual plot

has a slight downward trend rather than random scatter, some non-linear structure is still uncaptured. Training and validation MSE stay close until around $\alpha=10$ then rise together at high α , that is underfitting. The coefficient chart tells the clearest story, Property Age and Income swing wildly under plain Linear Regression (46 thousand either direction), Ridge crushes that down and Lasso all but erases it, while Loan Amount Request stays large and stable across every model.

11 Discussion

Regularization did not meaningfully beat the plain Linear Regression baseline on accuracy, both land around $R^2=0.60$ on the test set. What it clearly did deliver is more trustworthy coefficients, the wild swings in Property Age and Income under plain Linear Regression looked like overfitting to collinearity between those two features, and Ridge and Lasso both calm that down. Lasso zeroed 5 of 45 coefficients, mostly one-hot columns carrying little independent signal once the strong numeric predictors are already in the model. The residual pattern suggests a non-linear model would help more than further regularization tuning at this point.

12 Conclusion

Linear Regression already sits close to the practical ceiling of this feature set, test R^2 around 0.60, and Ridge/Lasso/Elastic Net mainly earn their keep through more stable coefficients rather than higher accuracy. Ridge $\alpha=100$ and Lasso $\alpha=10$ were both fairly strong settings, consistent with real collinearity in the data worth damping down.

13 Learning Outcomes

- Saw that regularization does not always move accuracy much, its real value here was stabilizing coefficients that were clearly overfitting under plain Linear Regression.
- Understood the practical difference between Ridge shrinking weights and Lasso zeroing them out.
- Learned to read a training vs validation error curve as a picture of underfitting and overfitting rather than just a definition.
- Noticed that GridSearchCV time can balloon when a model has convergence trouble at certain hyperparameter values, not just from the grid size itself.

14 References

1. Stephen Marsland, “Machine Learning - An Algorithmic Perspective”, 2nd Edition, Chapman and Hall/CRC Machine Learning and Pattern Recognition Series, 2015.
2. Ethem Alpaydin, “Introduction to Machine Learning”, 3rd Edition, The MIT Press, 2014.